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EULER GRAPHS

Graph theory was born in 1736 with Euler's famous paper in which he solved the Konigsberg bridge problem. In the same paper, Euler posed (and then solved) a more general problem: In what type of graph G is it possible to find a closed walk running through every edge of G

exactly once? Such a walk is now called an *Euler line*, and a graph that consists of an Euler line is called an *Euler graph*. More formally:

If some closed walk in a graph contains all the edges of the graph, then the walk is called an *Euler line* and the graph an *Euler graph*.

By its very definition a walk is always connected. Since the Euler line (which is a walk) contains all the edges of the graph, an *Euler graph* is always connected, except for any isolated vertices the graph may have. Since isolated vertices do not contribute anything to the understanding of an Euler graph, it is hereafter assumed that Euler graphs do not have any isolated vertices and are therefore connected.

Now we shall state and prove an important theorem, which will enable us to tell immediately whether or not a given graph is an Euler graph.

THEOREM 2-4

A given connected graph G is an Euler graph if and only if all vertices of G are of even degree.

Proof: Suppose that G is an Euler graph. It therefore contains an Euler line (which is a closed walk). In tracing this walk we observe that every time the walk meets a vertex v it goes through two "new" edges incident on v —with one we "entered" v and with the other "exited." This is true not only of all intermediate vertices of the walk but also of the terminal vertex, because we "exited" and "entered" the same vertex at the beginning and end of the walk, respectively. Thus if G is an Euler graph, the degree of every vertex is even.

To prove the sufficiency of the condition, assume that all vertices of G are of even degree. Now we construct a walk starting at an arbitrary vertex v and going through the edges of G such that no edge is traced more than once. We continue tracing as far as possible. Since every vertex is of even degree, we can exit from every vertex we enter; the tracing cannot stop at any vertex but v . And since v is also of even degree, we shall eventually reach v when the tracing comes to an end. If this closed walk h we just traced includes all the edges of G , G is an Euler graph. If not, we remove from G all the edges in h and obtain a subgraph h' of G formed by the remaining edges. Since both G and h have all their vertices of even degree, the degrees of the vertices of h' are also even. Moreover, h' must touch h at least at one vertex a , because G is connected. Starting from a , we can again construct a new walk in graph h' . Since all the vertices of h' are of even degree, this walk in h' must terminate at vertex a ; but this walk in h' can be combined with h to form a new walk, which starts and ends at vertex v and has

more edges than h . This process can be repeated until we obtain a closed walk that traverses all the edges of G . Thus G is an Euler graph..

Konigsberg Bridge Problem: Now looking at the graph of the Konigsberg bridges (Fig. 1-5), we find that not all its vertices are of even degree. Hence, it is not an Euler graph. Thus it is not possible to walk over each of the seven bridges exactly once and return to the starting point.

One often encounters Euler lines in various puzzles. The problem common to these puzzles is to find how a given picture can be drawn in one continuous line without retracing and without lifting the pencil from the paper. Two such pictures are shown in Fig. 2-12. The drawing in Fig. 2-12(a) is called *Mohammed's scimitars* and is believed to have come from the Arabs. The one in Fig. 2-12(b) is, of course, the *star of David*. (Equal time!)

In defining an Euler line some authors drop the requirement that the walk be closed. For example, the walk $a \rightarrow 1 \rightarrow c \rightarrow 2 \rightarrow d \rightarrow 3 \rightarrow a \rightarrow 4 \rightarrow b \rightarrow 5 \rightarrow d \rightarrow 6 \rightarrow e \rightarrow 7 \rightarrow b$ in Fig. 2-13, which includes all the edges of the graph and does not retrace any edge, is not closed. The initial vertex is a and the final vertex is b . We shall call such an open walk that includes (or traces or covers) all edges of a graph without retracing any edge a *unicursal line* or an *open Euler line*. A (connected) graph that has a unicursal line will be called a *unicursal graph*.

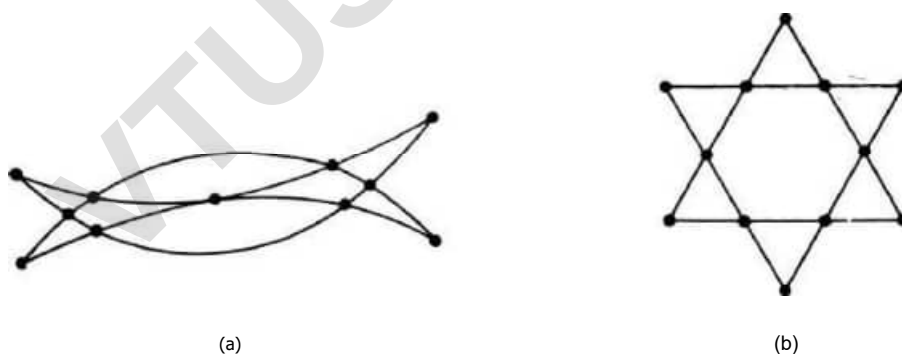


Fig. 2-12 Two Euler graphs.

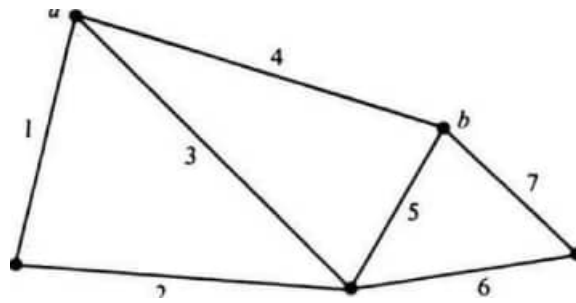


Fig. 2-13 Unicursal graph.

It is clear that by adding an edge between the initial and final vertices of a unicursal line we shall get an Euler line. Thus a connected graph is unicursal if and only if it has exactly two vertices of odd degree. This observation can be generalized as follows:

THEOREM 2-5

In a connected graph G with exactly $2k$ odd vertices, there exist k edge-disjoint subgraphs such that they together contain all edges of G and that each is a unicursal graph.

Proof: Let the odd vertices of the given graph G be named $v_1, v_2, \dots, v_k; w_1, w_2, \dots, w_k$ in any arbitrary order. Add k edges to G between the vertex pairs $(v_1, w_1), (v_2, w_2), \dots, (v_k, w_k)$ to form a new graph G' .

Since every vertex of G' is of even degree, G' consists of an Euler line p . Now if we remove from p the k edges we just added (no two of these edges are incident on the same vertex), p will be split into k walks, each of which is a unicursal line: The first removal will leave a single unicursal line; the second removal will split that into two unicursal lines; and each successive removal will split a unicursal line into two unicursal lines, until there are k of them. Thus the theorem. $\square$

We shall interrupt our study of Euler graphs to define some commonly used graph-theoretic operations. One of these operations is required immediately in the next section; others will be needed later.

OPERATIONS ON GRAPHS

As is the case with most mathematical entities, it is convenient to consider a

large graph as a combination of small ones and to derive its properties from those of the small ones. Since graphs are defined in terms of the sets of vertices and edges, it is natural to employ the set-theoretical terminology to define operations between graphs. In particular:

The *union* of two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ is another graph G_3 (written as $G_3 = G_1 \cup G_2$) whose vertex set $V_3 = V_1 \cup V_2$ and the edge set $E_3 =$

$E_1 \cup E_2$. Likewise, the *intersection* $G_1 \cap G_2$ of graphs G_1 and G_2 is a graph G_4 consisting only of those vertices and edges that are in both G_1 and G_2 . The *ring sum* of two graphs G_1 and G_2 (written as $G_1 \oplus G_2$) is a graph consisting of the vertex set $V_1 \cup V_2$ and of edges that are either in G_1 or G_2 , but *not* in both. Two graphs and their union, intersection, and ring sum are shown in Fig. 2-14.1.

It is obvious from their definitions that the three operations just mentioned are commutative. That is,

$$\begin{aligned} G_1 \cup G_2 &= G_2 \cup G_1 & G_1 \cap G_2 &= G_2 \cap G_1, \\ G_1 \oplus G_2 &= G_2 \oplus G_1. \end{aligned}$$

If G_1 and G_2 are edge disjoint, then $G_1 \cap G_2$ is a null graph, and $G_1 \oplus G_2 = G_1 \cup G_2$. If G_1 and G_2 are vertex disjoint, then $G_1 \cap G_2$ is empty. For any graph G ,

$$G \cup G = G \cap G = G,$$

and

$$G \oplus G = \text{a null graph.}$$

If g is a subgraph of G , then $G \oplus g$ is, by definition, that subgraph of G which remains after all the edges in g have been removed from G . Therefore, $G \oplus g$ is written as $G - g$, whenever $g \subset G$. Because of this complementary nature, $G \oplus g = G - g$ is often called the complement of g in G .

Decomposition: A graph G is said to have been *decomposed* into two subgraphs g_1 and g_2 if

$$g_1 \cup g_2 = G,$$

and

$$g_1 \cap g_2 = \text{a null graph.}$$

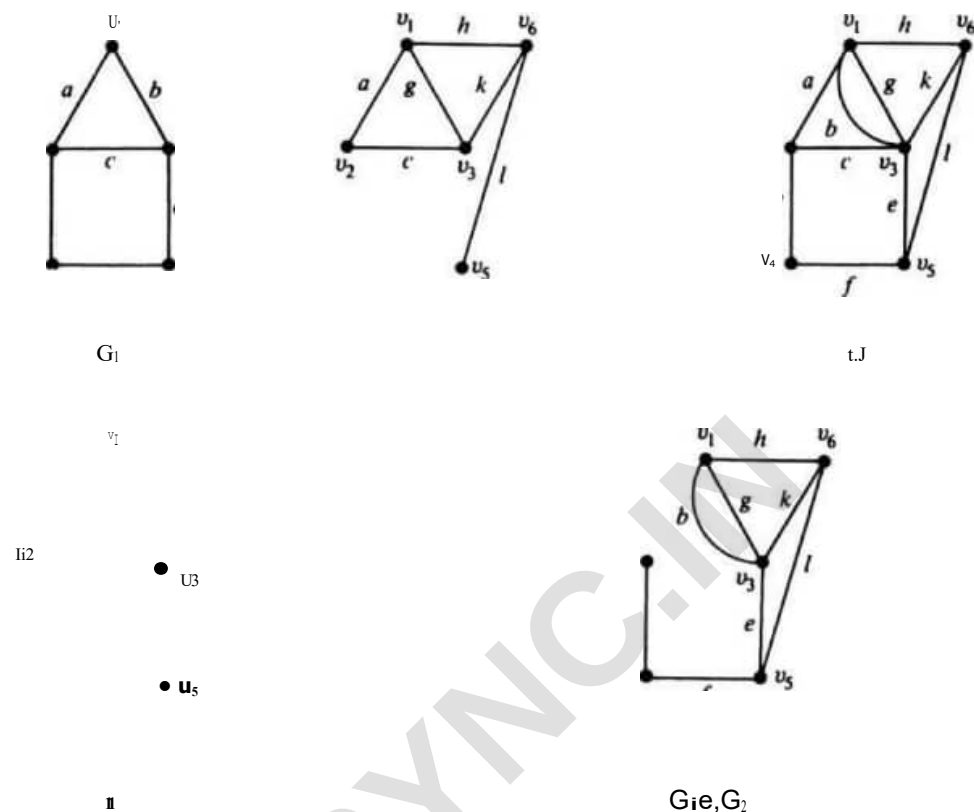


Fig. 2-14 Union, intersection, and ring sum of two graphs.

In other words, every edge of G occurs either in g_1 or in g_2 , but not in both. Some of the vertices, however, may occur in both g_1 and g_2 . In decomposition, isolated vertices are disregarded. A graph containing m edges $\{e_1, e_2, \dots, e_m\}$ can be decomposed in $2^m - 1$ different ways into pairs of subgraphs g_1, g_2 (why?).

Although union, intersection, and ring sum have been defined for a pair of graphs, these definitions can be extended in an obvious way to include any finite number of graphs. Similarly, a graph G can be decomposed into more than two subgraphs—subgraphs that are (pairwise) edge disjoint and collectively include every edge in G .

Deletion: If v_i is a vertex in graph G , then $G - v_i$ denotes a subgraph of G obtained by deleting (i.e., removing) v_i from G . Deletion of a vertex always implies the deletion of all edges incident on that vertex. (See Fig. 2-15.) If e_i is an edge in G , then $G - e_i$ is a subgraph of G obtained by deleting e_i from G . Deletion of an edge does not imply deletion of its end vertices. Therefore $G - e_i$

$= G - e.$

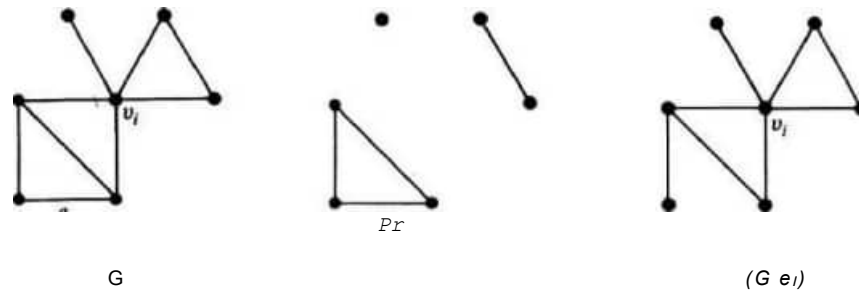


Fig. 2-15 Vertex deletion and edge deletion.

Fusion: A pair of vertices a, b in a graph are said to be *fused (merged or identified)* if the two vertices are replaced by a single new vertex such that every edge that was incident on either a or b or on both is incident on the new vertex. Thus fusion of two vertices does not alter the number of edges, but it reduces the number of vertices by one. See Fig. 2-16 for an example.

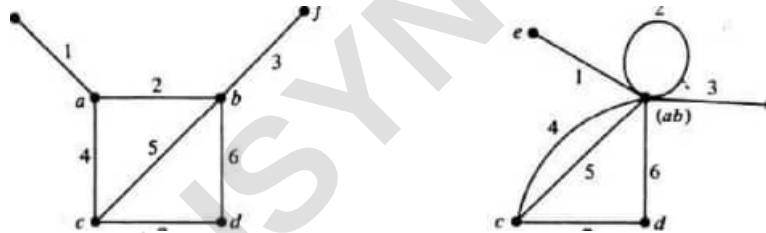


Fig. 2-16 Fusion of vertices a and b .

These are some of the elementary operations on graphs. More complex operations have been defined and are used in graph-theory literature. For a survey of such operations see the paper by Harary and Wilcox [2-10].

MORE ON EULER GRAPHS

The following are some more results on the important topic of Euler graphs.

THEOREM 2-6

A connected graph G is an Euler graph if and only if it can be decomposed into circuits.

Proof: Suppose graph G can be decomposed into circuits; that is, G is a union

of edge-disjoint circuits. Since the degree of every vertex in a circuit is two, the degree of every vertex in G is even. Hence G is an Euler graph.

Conversely, let G be an Euler graph. Consider a vertex v_1 . There are at least two edges incident at v_1 . Let one of these edges be between v_1 and v_2 . Since vertex v_2 is also of even degree, it must have at least another edge, say between v_2 and v_3 . Proceeding in this fashion, we eventually arrive at a vertex that has previously been traversed, thus forming a circuit F . Let us remove F from G . All vertices in the remaining graph (not necessarily connected) must also be of even degree. From the remaining graph remove another circuit in exactly the same way as we removed F from G . Continue this process until no edges are left. Hence the theorem. ■

Arbitrarily Traceable Graphs: Consider the graph in Fig. 2-17, which is an Euler graph. Suppose that we start from vertex a and trace the path $a b c$.



Fig. 2-17 Arbitrarily traceable graph from c .

Now at c we have the choice of going to a , d , or e . If we took the first choice, we would only trace the circuit $a b c a$, which is not an Euler line. Thus, starting from a , we cannot trace the entire Euler line simply by moving along any edge that has not already been traversed. This raises the following interesting question:

What property must a vertex v in an Euler graph have such that an Euler line is always obtained when one follows any walk from vertex v according to the single rule that whenever one arrives at a vertex one shall select *any* edge (which has not been previously traversed)?

Such a graph is called an *arbitrarily traceable graph from vertex v* . For instance, the Euler graph in Fig. 2-17 is an arbitrarily traceable graph from vertex c , but not from any other vertex. The Euler graph in Fig. 2-18 is not arbitrarily traceable from any vertex; the graph in Fig. 2-19 is arbitrarily traceable from all its vertices. The following interesting theorem, due to Ore [2-5], answers the question just raised.



Fig. 2-18 Euler graph; not arbitrarily traceable.

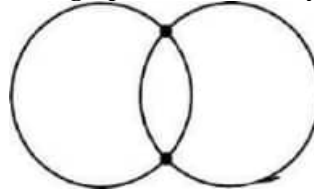


Fig. 2-19 Arbitrarily traceable graph from all vertices.

THEOREM 2-7

An Euler graph G is arbitrarily traceable from vertex v in G if and only if every circuit in G contains v .

For a proof of the theorem the reader is referred to [2-5].

HAMILTONIAN PATHS AND CIRCUITS

An Euler line of a connected graph was characterized by the property of being a closed walk that traverses *every edge* of the graph (exactly once). A *Hamiltonian circuit* in a connected graph is defined as a closed walk that traverses *every vertex* of G exactly once, except of course the starting vertex, at which the walk also terminates. For example, in the graph of Fig. 2-20(a) starting at vertex v , if one traverses along the edges shown in heavy lines—passing through each vertex exactly once—one gets a Hamiltonian circuit. A Hamiltonian circuit for the graph in Fig. 2-20(b) is also shown by heavy lines. More formally:

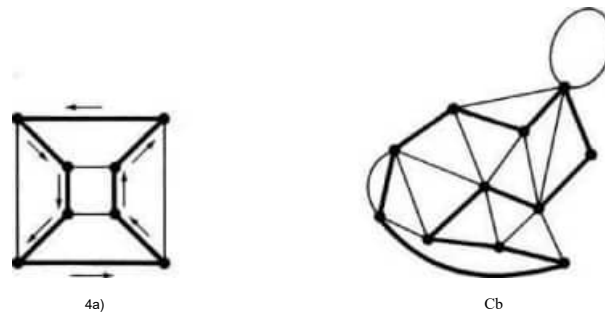


Fig. 2-20 Hamiltonian circuits.

A circuit in a connected graph G is said to be Hamiltonian if it includes every vertex of G . Hence a Hamiltonian circuit in a graph of n vertices consists of exactly n edges.

Obviously, not every connected graph has a Hamiltonian circuit. For example, neither of the graphs shown in Figs. 2-17 and 2-18 has a Hamiltonian circuit. This raises the question: What is a necessary and sufficient condition for a connected graph G to have a Hamiltonian circuit?

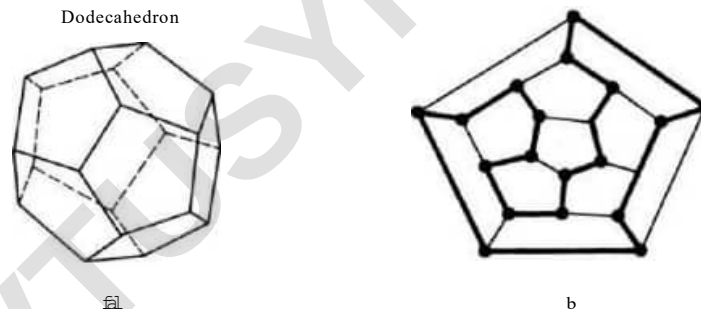


Fig. 2-21 Dodecahedron and its graph shown with a Hamiltonian circuit.

This problem, first posed by the famous Irish mathematician Sir William Rowan Hamilton in 1859, is still unsolved. As was mentioned in Chapter 1, Hamilton made a regular dodecahedron of wood [see Fig. 2-21(a)], each of whose 20 corners was marked with the name of a city. The puzzle was to start from any city and find a route along the edge of the dodecahedron that passes through every city exactly once and returns to the city of origin. The graph of the dodecahedron is given in Fig. 2-21(b), and one of many such routes (a Hamiltonian circuit) is shown by heavy lines.

The resemblance between the problem of an Euler line and that of a Hamiltonian circuit is deceptive. The latter is infinitely more complex. Although one can find Hamiltonian circuits in many specific graphs, such as those shown

in Figs. 2-20 and 2-21, there is no known criterion we can apply to determine the existence of a Hamiltonian circuit in general. There are, however, certain types of graphs that always contain Hamiltonian circuits, as will be presently shown.

Hamiltonian Path: If we remove any one edge from a Hamiltonian circuit, we are left with a path. This path is called a *Hamiltonian path*. Clearly, a Hamiltonian path in a graph G traverses every vertex of G . Since a Hamiltonian path is a subgraph of a Hamiltonian circuit (which in turn is a subgraph of another graph), every graph that has a Hamiltonian circuit also has a Hamiltonian path. There are, however, many graphs with Hamiltonian paths that have no Hamiltonian circuits (Problem 2-23). The length of a Hamiltonian path (if it exists) in a connected graph of n vertices is $n - 1$.

In considering the existence of a Hamiltonian circuit (or path), we need only consider simple graphs. This is because a Hamiltonian circuit (or path) traverses every vertex exactly once. Hence it cannot include a self-loop or a set of parallel edges. Thus a general graph may be made simple by removing parallel edges and self-loops before looking for a Hamiltonian circuit in it.

It is left as an exercise for the reader to show that neither of the two graphs shown in Fig. 2-22 has a Hamiltonian circuit (or Hamiltonian path). See Problem 2-24.

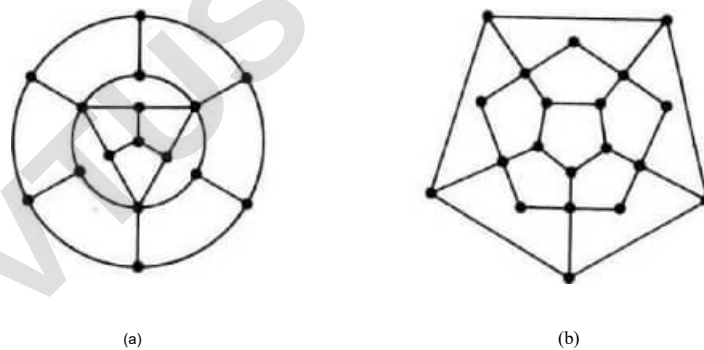


Fig. 2.22 Graphs without Hamiltonian circuits.

What general class of graphs is guaranteed to have a Hamiltonian circuit? Complete graphs of three or more vertices constitute one such class.

Complete Graph: A simple graph in which there exists an edge between every pair of vertices is called a *complete graph*. Complete graphs of two, three, four, and five vertices are shown in Fig. 2-23. A complete graph is sometimes also referred to as a *universal graph* or a *clique*. Since every vertex is joined with every other vertex through one edge, the degree of every vertex is $n - 1$ in a

complete graph G of n vertices. Also the total number of edges in G is $n(n - 1)/2$. See [Problem 1-12](#).



Fig. 2-23 Complete graphs of two, three, four, and five vertices.

It is easy to construct a Hamiltonian circuit in a complete graph of n vertices. Let the vertices be numbered $v_1, v_2, \dots, v_n$. Since an edge exists between any two vertices, we can start from v_1 and traverse to v_2 , and v_3 , and so on to v_n , and finally from v_n to v_1 . This is a Hamiltonian circuit.

Number of Hamiltonian Circuits in a Graph: A given graph may contain more than one Hamiltonian circuit. Of interest are all the edge-disjoint Hamiltonian circuits in a graph. The determination of the exact number of *edge-disjoint* Hamiltonian circuits (or paths) in a graph in general is also an unsolved problem. However, the number of edge-disjoint Hamiltonian circuits in a complete graph with odd number of vertices is given by [Theorem 2-8](#).

THEOREM 2-8

In a complete graph with n vertices there are $(n - 1)/2$ edge-disjoint Hamiltonian circuits, if n is an odd number ≥ 3 .

Proof: A complete graph G of n vertices has $n(n - 1)/2$ edges, and a Hamiltonian circuit in G consists of n edges. Therefore, the number of edge-disjoint Hamiltonian circuits in G cannot exceed $(n - 1)/2$. That there are $(n - 1)/2$ edge-disjoint Hamiltonian circuits, when n is odd, can be shown as follows:

The subgraph (of a complete graph of n vertices) in [Fig. 2-24](#) is a Hamiltonian circuit. Keeping the vertices fixed on a circle, rotate the polygonal pattern clockwise by $360/(n - 1)$, $2 \cdot 360/(n - 1)$, $3 \cdot 360/(n - 1)$, $\dots$, $(n - 3) \cdot 360/(n - 1)$ degrees. Observe that each rotation produces a Hamiltonian circuit that has no edge in common with any of the previous ones. Thus we have $(n - 1)/2$ new Hamiltonian circuits, all edge disjoint from the one in [Fig. 2-24](#) and also edge disjoint among themselves. Hence the theorem. ■

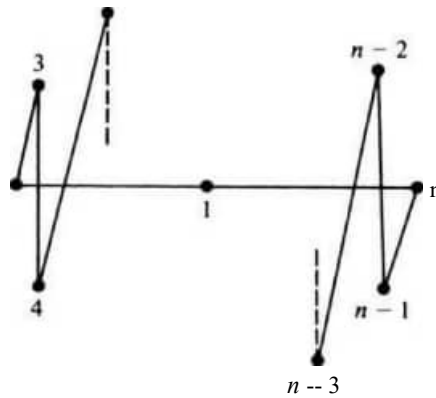


Fig. 2-24 Hamiltonian circuit; n is odd.

This theorem enables us to solve the problem of the seating arrangement at a round table, introduced in [Chapter 1](#), as follows:

Representing a member x by a vertex and the possibility of his sitting next to another member y by an edge between x and y , we construct a graph G . Since every member is allowed to sit next to any other member, G is a complete graph of nine vertices—nine being the number of people to be seated around the table. Every Seating arrangement around the table is clearly a Hamiltonian circuit.

The first day of their meeting they can sit in any order, and it will be a Hamiltonian circuit H_1 . The second day, if they are to sit such that every member must have different neighbors, we have to find another Hamiltonian circuit H_2 in G , with an entirely different set of edges from those in H_1 ; that is, H_1 and H_2 are edge-disjoint Hamiltonian circuits. From [Theorem 2-8](#) the number of edge-disjoint Hamiltonian circuits in G is four; therefore, only four such arrangements exist among nine people.

Another interesting result on the question of existence of Hamiltonian circuits in a graph, obtained by G. A. Dirac, is:

THEOREM 2-9

A sufficient (but by no means necessary) condition for a simple graph G to have a Hamiltonian circuit is that the degree of every vertex in G be at least $n/2$, where n is the number of vertices in G .

Proof: For proof the reader is referred to the original paper by Dirac [2-3].

TRAVELING-SALESMAN PROBLEM

A problem closely related to the question of Hamiltonian circuits is the *traveling-salesman problem*, stated as follows: A salesman is required to visit a number of cities during a trip. Given the distances between the cities, in what order should he travel so as to visit every city precisely once and return home, with the minimum mileage traveled?

Representing the cities by vertices and the roads between them by edges, we get a graph. In this graph, with every edge e_i there is associated a real number (the distance in miles, say), $w(e_i)$. Such a graph is called a *weighted graph*; $w(e_i)$ being the *weight* of edge e_i .

In our problem, if each of the cities has a road to every other city, we have a *complete weighted graph*. This graph has numerous Hamiltonian circuits, and we are to pick the one that has the smallest sum of distances (or weights).

The total number of different (not edge disjoint, of course) Hamiltonian circuits in a complete graph of n vertices can be shown to be $(n - 1)!/2$. This follows from the fact that starting from any vertex we have $n - 1$ edges to choose from the first vertex, $n - 2$ from the second, $n - 3$ from the third, and so on. These being independent choices, we get $(n - 1)!$ possible number of choices. This number is, however, divided by 2, because each Hamiltonian circuit has been counted twice.

Theoretically, the problem of the traveling salesman can always be solved by enumerating all $(n - 1)!/2$ Hamiltonian circuits, calculating the distance traveled in each, and then picking the shortest one. However, for a large value of n , the labor involved is too great even for a digital computer (try solving it for the 50 state capitals in the United States; $n = 50$).

The problem is to prescribe a manageable algorithm for finding the shortest route. No efficient algorithm for problems of arbitrary size has yet been found, although many attempts have been made. Since this problem has applications in operations research, some specific large-scale examples have been worked out (see [2-1]). There are also available several heuristic methods of solution that give a route very close to the shortest one, but do not guarantee the shortest (see [2-4] for such a method).

SUMMARY

In this chapter we discussed the subgraph—a graph that is part of another

graph. Walks, paths, circuits, Euler lines, Hamiltonian paths, and Hamiltonian circuits in a graph G are its subgraphs with special properties. A given graph G can be characterized and studied in terms of the presence or absence of these subgraphs. Many physical problems can be represented by graphs and solved by observing the relevant properties of the corresponding graphs.

Various types of walks discussed in this chapter are summarized in Fig. 2-25. The arrows point in the direction of increasing restriction.

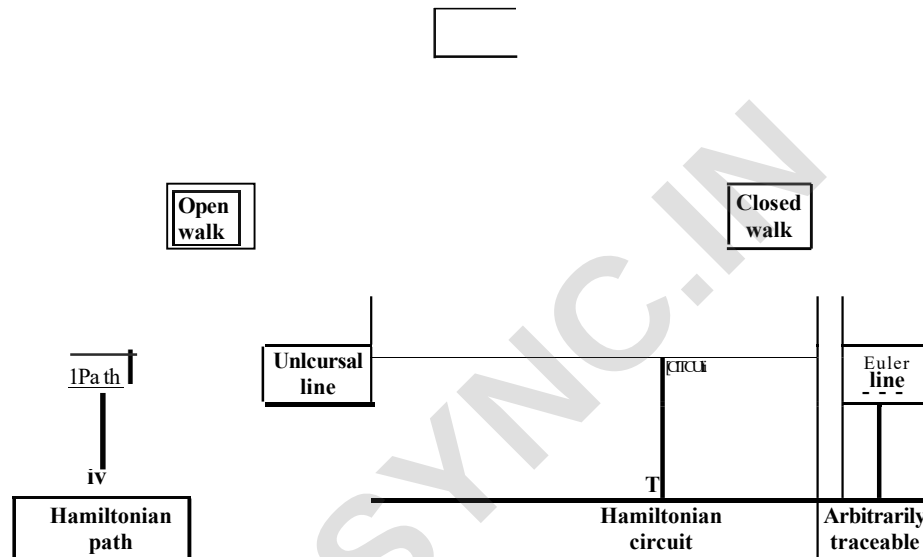


Fig. 2-25 Different types of walks.

REFERENCES

Textbooks listed in Chapter 1 are to be read for this chapter also. Specially recommended are

1. Berge [1-1], Chapters 1, 7, 11, and 17.
2. Busacker and Saaty [1-2], Chapters 1, 3, and 6.
3. Harary [1-5], Chapter 7.
4. Ore [1-9], Chapters 2 and 3.
5. Ore [1-10], Chapters 1 and 2.
6. Seshu and Reed [1-13], Chapter 2.

For arbitrarily traceable graphs, one should read Ore's paper [2-5]. Additional

information on properties of Hamiltonian graphs can be found in papers by Tutte [2-8], Ore [2-6], Smith and Tutte [2-7], and Dirac [2-3]. Chapters 4 and 5 of Tutte's book [2-9] are also devoted to paths and Euler paths. On the traveling-salesman problem there are many papers. In an excellent survey Bellmore and Nemhauser [2-1] summarize and list most of these papers. Deo and Hakimi [2-2] generalized the Hamiltonian-path problem and applied it to a wiring problem in computers.

- 2-1. BELLMORE, M., and G. L. NEMHAUSER, "The Traveling Salesman Problem: A Survey," *Operations Res.*, Vol. 16, 1968, 538-558.
- 2-2. DEO, N., and S. L. HAKIMI, "The Shortest Generalized Hamiltonian Tree," *Proc. Third Annual Allerton Conf.*, University of Illinois, 1965, 879-888.
- 2-3. DIRAC, G. A., "Connectivity Theorems for Graphs," *Quart J. Math. Oxford*, Ser. (2), Vol. 3, 1952, 171-174.
- 2-4. LIN, S., "Computer Solution of the Traveling Salesman Problem," *BSTJ*, Vol. 44, 1965, 2245-2269.
- 2-5. ORE, O., "A Problem Regarding the Tracing of Graphs," *Rev. Elementary Math.*, .6, 1961, 49-53.
- 2-6. ORE, O., "Note on Hamilton Circuits," *Am. Math. Monthly*, Vol. 67, 1960, 55.
- 2-7. SMITH, C. A. B., and W. T. TUTTE, "On Unicursal Paths in a Network of Degree Four," *Am. Math. Monthly*, Vol. 48, 1941, 233-237.
- 2-8. TUTTE, W. T., "On Hamiltonian Circuits," *J. London Math. Soc.*, Vol. 21, 1946, 98-101.
- 2-9. TUTTE, W. T., *Connectivity in Graphs*, University of Toronto Press, Toronto, 1966.
- 2-10. HARARY, F., and G. W. WILCOX, "Boolean Operations on Graphs," *Math. Scand.*, Vol. 20, 1967, 41-51.

PROBLEMS

- 2-1. Verify that the two graphs in Fig. 2-2 are isomorphic. Label the corresponding vertices and edges.
- 2-2. Show by redrawing, step by step, that graphs (b) and (c) in Fig. 2-3. are isomorphic to (a).
- 2-3. Show that the two graphs in Figs. 2-26(a) and (b) are isomorphic.

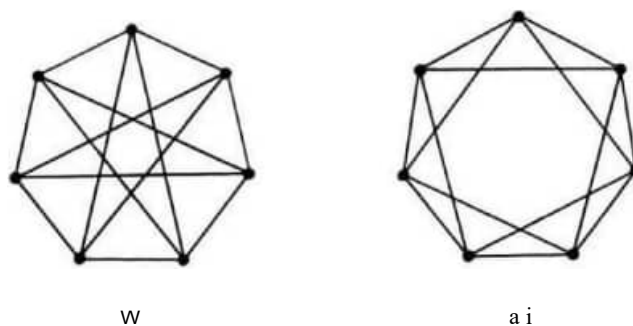


Fig. 2-26

- 2-4. Construct three more examples to show that conditions 1, 2, and 3 in [Section 2-1](#) are not sufficient for isomorphism between graphs.
- 2-5. Prove that any two simple connected graphs with n vertices, all of degree two, are isomorphic.
- 2-6. Are the two graphs in [Fig. 2-27](#) isomorphic? Why?

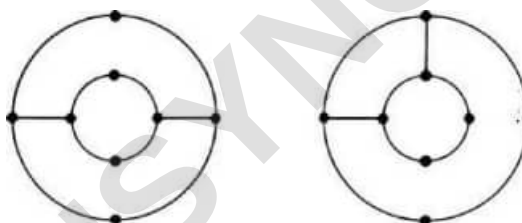


Fig. 2-27

- 2-7. Given the set of cubes represented by the graph in [Fig. 2-6](#), is it possible to stack all four cubes into a column such that each side shows only one color? Explain.
- 2-8. Prove that a simple graph with n vertices must be connected if it has more than $\frac{1}{2}(n-1)(n-2)$ edges. (*Hint: Use [Theorem 2-3](#).*)
- 2-9. Prove that if a connected graph G is decomposed into two subgraphs g_1 and g_2 , there must be at least one vertex common between g_1 and g_2 .
- 2-10. Prove that a connected graph G remains connected after removing an edge e_i from G , if and only if e_i is in some circuit in G .
- 2-11. Draw a connected graph that becomes disconnected when any edge is removed from it.
- 2-12. Prove that a graph with n vertices satisfying the condition of [Problem 2-11](#) is (a) simple, and (b) has exactly $n-1$ edges.
- 2-13. What is the length of the path from the entrance to the center of the maze in [Problem 1-7](#)?

- 2-14. List all the different paths between vertices 5 and 6 in [Fig. 2-5\(a\)](#). Give the length of each of these paths.
- 2-15. Group the paths listed in [Problem 2-14](#) into sets of edge-disjoint paths. Demonstrate that the union of two edge-disjoint paths between a pair of vertices forms a circuit.
- 2-16. In a graph G let p_i and p_j be two different paths between two given vertices. Prove that $p_i \oplus p_j$ is a circuit or a set of circuits in G .
- 2-17. Let a , b , and c be three distinct vertices in a graph. There is a path between a and b and also there is a path between b and c . Prove that there is a path between a and c .
- 2-18. If the intersection of two paths is a disconnected graph, show that the union of the two paths has at least one circuit.
- 2-19. You are given a 10-piece domino set whose tiles have the following set of dots: (1, 2); (1, 3); (1, 4); (1, 5); (2, 3); (2, 4); (2, 5); (3, 4); (3, 5); (4, 5). Discuss the possibility of arranging the tiles in a connected series such that one number on a tile always touches the same number on its neighbor. (*Hint: Use a five-vertex complete graph and see if it is an Euler graph.*)
- 2-20. Is it possible to move a knight on a chessboard such that it completes every permissible move exactly once? A move between two squares is counted as one regardless of the direction in which it is made. (*Hint: Is the graph of [Problem 1-6](#) unicursal?*)
- 2-21. A round-robin tournament (when every player plays against every other) among n players (n being an even number) can be represented by a complete graph of n vertices. Discuss how you would schedule the tournaments to finish in the shortest possible time.
- 2-22. Observe that there can be no path longer than a Hamiltonian path (if it exists) in a graph.
- 2-23. Draw a graph that has a Hamiltonian path but does not have a Hamiltonian circuit.
- 2-24. Show that neither of the graphs in [Fig. 2-22](#) has a Hamiltonian path (and therefore no Hamiltonian circuit). [*Hint: For [Fig. 2-22\(a\)](#), of all the edges incident at a vertex only two can be included in a Hamiltonian circuit. Count the number of edges that have to be excluded. You will find that 13 edges must be excluded from [Fig. 2-22\(a\)](#). The number of remaining edges is insufficient to form a Hamiltonian circuit. For [Fig. 2-22\(b\)](#), first consider all vertices of degree two.*]
- 2-25. Show that the graph of a rhombic dodecahedron (with eight vertices of

degree three and six vertices of degree four) has no Hamiltonian path (and therefore no Hamiltonian circuit).

2-26. Draw a graph in which an Euler line is also a Hamiltonian circuit. What can you say about such graphs in general?

2-27. Is it possible, starting from any of the 64 squares of the chessboard, to move a knight such that it occupies every square exactly once and returns to the initial position? If so, give one such tour. (*Hint: Look for a Hamiltonian circuit in the graph of Problem 1-6.*)

2-28. Prove that a graph G with n vertices always has a Hamiltonian path if the sum of the degrees of every pair of vertices v_i, v_j in G satisfies the condition

$$d(v_i) + d(v_j) \geq n - 1.$$

(*Hint: First show that G is connected. Then use induction on path length in G .*)

2-29. Using the result of Problem 2-28, show that in a dancing ring of n children it is always possible to arrange the children so that everyone has a friend at each side if every child enjoys friendship with at least half the children.

For brevity, a vertex with odd degree is called an *odd vertex*, and a vertex with even degree an *even vertex*.
Proof: Let G be a graph with n vertices and k edges. Squaring both sides,

$$\sum_{i=1}^n (d(v_i) - 1)^2 = \sum_{i=1}^n d(v_i)^2 - 2 \sum_{i=1}^n d(v_i) + n = \sum_{i=1}^n d(v_i)^2 - 2k + n$$

Since $\sum_{i=1}^n d(v_i) = 2k$, the right-hand side is $\sum_{i=1}^n d(v_i)^2 - 2k + n$. Because $(d(v_i) - 1)^2 \geq 0$ for all i , we have

if an edge ei is in two graphs G_1 and G_2 , its end vertices in G_1 must have the same labels as in G_2 .